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Research Article

Synergistic enhancement of electrochemical performance in lignin-based carbon aerogel supercapacitors through phytic acid-induced spherical structure formation and dual P/S heteroatom doping

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ABSTRACT

The widespread deployment of renewable energy sources worldwide, such as wind power and photovoltaics, has created an urgent need for efficient energy storage systems. Biomass-derived carbon aerogels, due to their environmentally friendly and sustainable properties, have emerged as ideal precursor materials for advanced energy storage applications, particularly in supercapacitors. This study developed a method to prepare phytate-induced phosphorus/sulfur (P/S) co-doped magnesium lignosulfonate-based carbon aerogels (LCAs). Phytate induction facilitated the formation of regular spherical structures while simultaneously optimizing surface morphology and enabling efficient and uniform doping of P and S heteroatoms. The optimized sample, LCA-2-700, the lignin-based carbon aerogel that was prepared with the magnesium lignosulfonate (LS):sodium alginate (SA):phytic acid (PA) mass ratio of 5:5:2 and carbonized at 700 °C, exhibited a specific capacitance of 362 F/g at the current density of 0.5 A/g, with the assembled device achieving an energy density of 40.1 W·h/kg at a power density of 700 W/kg. After 20,000 cycles, the capacitance retention rate remained at 82.5%, demonstrating excellent electrochemical durability. The high performance was attributed to the synergistic effects of its spherical structure, high specific surface area, and P/S dual-heteroatom doping. This study provides an effective approach for synergistic structure-doping regulation of lignin-based carbon aerogels and provides a potential pathway for practical applications in high-performance supercapacitors.

1. Introduction

The escalating global energy crisis and associated environmental concerns have intensified research efforts toward renewable energy and advanced energy storage systems (Fu et al., 2017; Li et al., 2025; Wang et al., 2026). Supercapacitors (SCs) are at the forefront of energy storage research owing to their high-power density, substantial energy density, long-term cycling stability,

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In the three-electrode system, all tests were performed in an aqueous solution of 6 mol/L KOH with Hg/HgO as the reference electrode and a Pt wire as the counter electrode. Cyclic voltammetry (CV) and galvanostatic charge-discharge (GCD) tests were performed in a potential window from –1 to 0 V.

As a two-electrode system, two AC electrodes with the same mass load were placed in the battery. An aqueous solution of 6 mol/L KOH was used as the electrolyte. The electrochemical performance of the symmetric supercapacitor (SSC) was tested in the voltage range from 0 to 1V. Electrochemical impedance spectroscopy (EIS) was performed from 10 mHz to 100 kHz at the open circuit voltage with an alternate current amplitude of 5 mV.

In the three-electrode system, the mass specific capacitance (C_s, F/g) of each AC was calculated from the recorded discharge curve according to Equation (1):

$$C_s = I\Delta t / m\Delta V_1 \tag{1}$$

where I, Δt, m, and ΔV_1 represent the constant discharge current (mA), discharge time (s), weight of the active material in the working electrode (mg), and voltage variation during discharge (V), respectively (Gao et al., 2024).

A symmetrical supercapacitor device was assembled by using two identical electrodes in 6 mol/L KOH to further investigate the electrochemical performances of carbon aerogels, and the corresponding parameters were calculated as shown in equations (2)–(4):

$$C_{cell} = I\Delta t / m_{total}\Delta V_2 \tag{2}$$

$$E = C_{cell}\Delta V_2^{\,2}/7.2 \tag{3}$$

$$P = 3600E/\Delta t \tag{4}$$

where C_{cell}, m_{total}, ΔV_2 Δt, E, and P represent the specific capacitance of the device (F/g), total mass of the active material in the working electrode (mg), potential change after the IR drop (V), discharge time corresponding to the voltage (s), energy density (W·h/kg), and power density (W/kg), respectively (Li et al., 2023).

3. Results and discussion

Fig. 2a displays a digital photograph of the pristine LCA-2 (the mass ratio of magnesium lignosulfonate:SA:PA = 5:5:2 and without carbonization) positioned on a flower. The absence of visible deformation in the flower underscores the highly lightweight characteristic of the aerogel. Fig. 2b presents digital images of both LCA-2 and LCA-2-700 (mass ratios of magnesium lignosulfonate:SA:PA = 5:5:2, and carbonized at 700 °C), illustrating that the carbonized sample LCA-2-700 retains its three-dimensional structural integrity without collapse, indicating good thermo-stability.

Fig. 2c shows the Fourier transform infrared (FT-IR) spectra of the non-carbonized lignin-based aerogel with the LS:SA mass ratio of 5:5 (LCA) and the non-carbonized lignin-based aerogel with the LS:SA:PA mass ratio of 5:5:2 (LCA-2). Both aerogels exhibited several common characteristic absorption bands. The broad band around 3500 cm^{-1} was assigned to O–H stretching vibrations, while the peak at 2910 cm^{-1} corresponded to aliphatic C–H stretching. The bands observed at 1628 and 1425 cm^{-1} were attributed to the

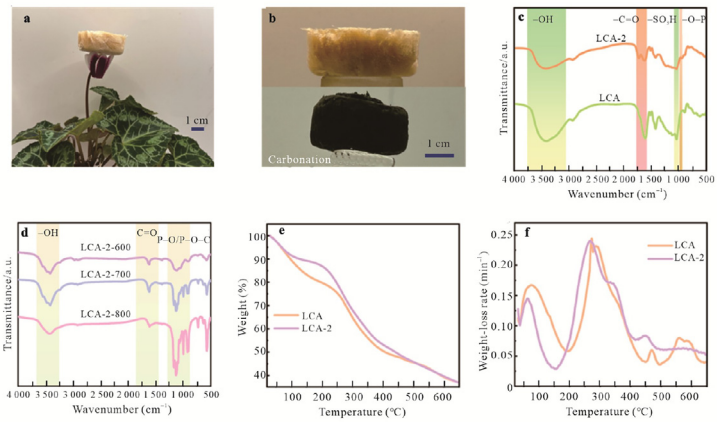

Fig. 2. (a) Digital photo of non-carbonized lignin-based aerogel with the magnesium lignosulfonate (LS):sodium alginate (SA):phytic acid (PA) mass ratio of 5:5:2 (LCA-2); (b) Digital photos of LCA-2 and lignin-based aerogel with LS:SA:PA mass ratio of 5:5:2 and carbonized at 700 °C (LCA-2-700); (c) Fourier transform infrared spectra of non-carbonized lignin-based aerogel with the LS:SA mass ratio of 5:5 (LCA) and LCA-2; (d) Fourier infrared spectra of lignin-based aerogel with LS:SA:PA mass ratio of 5:5:2 and carbonized at 600 °C (LCA-2-600), LCA-2-700, and lignin-based aerogel with LS:SA:PA mass ratio of 5:5:2 and carbonized at 800 °C (LCA-2-800); (e) Thermogravimetric analysis (TGA) and (f) differential thermo gravimetry (DTG) curves of LCA and LCA-2.

asymmetric and symmetric stretching modes of carboxylate groups, respectively. The absorption at 1733 cm^{-1} was associated with the C=O stretching vibration. Additionally, the band at 1023 cm^{-1} arose from C–O stretching. The absorptions at 1628 and 1516 cm^{-1} were representative of the aromatic skeleton vibration from lignin, and the band at 1043 cm^{-1} was assigned to the sulfonic acid group (–SO_3H) stretching. After the incorporation of PA, the peak at approximately 1030 cm^{-1} overlaped with lignin-derived features, which was consistent with asymmetric stretching of P–O–C, confirming successful phosphorus doping. Moreover, the introduction of PA enhanced intramolecular and intermolecular hydrogen bonding, leading to shifts of certain absorption bands toward either higher or lower wavenumbers.

The functional groups present in the carbon aerogel were further characterized by FT-IR spectroscopy as shown in Fig. 2d. The absorption bands between 1696 and 1635 cm^{-1} were assigned to stretching vibrations of the aromatic benzene skeleton. Additionally, the peaks located at 1120 and 995 cm^{-1} corresponded to P–O and P–O–C stretching vibrations, respectively, indicating the successful incorporation of phosphorus into the carbon framework after carbonization. During the carbonisation process of magnesium lignosulphonate at temperatures of 600–800 °C, the –SO_3H group underwent almost complete decomposition.

The pyrolytic behavior of LCA at the elevated temperatures was evaluated by thermogravimetric analysis. As depicted in Figs. 2e and 2f, the mass loss during heating from ambient temperature to 650 °C occurred in two distinct stages. The first stage, between room temperature and 200 °C, showed a mass loss of approximately 20%, primarily due to the evaporation of adsorbed water. This hygroscopicity originated from the abundance of carboxyl and hydroxyl groups in magnesium lignosulfonate and SA. As the temperature reached 100 °C, the release of absorbed water and volatile components occurred. The second mass loss stage, observed between 200 and 450 °C, exhibited a pronounced peak rate at around 275 °C, which was associated with the decomposition of magnesium lignosulfonate and SA, leading to the formation of a stable carbonaceous matrix. It was also found that the introduction of PA played a key protective role during pyrolysis. Owing to the corrosion resistance of PA and the release of low-boiling-point molecules, the initial decomposition temperature of the aerogel was reduced, thereby accelerating the early thermal degradation process and enhancing the overall thermal stability. Furthermore, the presence of PA facilitated the consolidation of the carbon architecture, which was critical for constructing a robust aerogel framework and provided a solid basis for its potential application in high-temperature environments.

Figs. 3a–3d present the FE-SEM images of LCA-700, LCA-1-700, LCA-2-700, and LCA-3-700. It could be observed that the surface of LCA-700 exhibited a rough and disordered morphology. With the incorporation of PA, the materials progressively develop a well-defined spherical architecture. The formation of spherical structures upon PA addition could be attributed to the coordination between PA and metal ions in LCA, which promoted self-assembly during gelation. The amount of PA influenced the size and uniformity of the spheres, as evidenced by FE-SEM images (Figs. 3a–3d). Based on the specific capacitance data for different carbon aerogels provided in Table S1 and Fig. S1, where LCA-2-700 exhibited the highest specific capacitance, we hypothesised that its outstanding electrochemical performance was closely linked to its pronounced spherical layered structure. When the temperature increased to 800 °C (Fig. 3f), the spherical architecture was retained but appeared to undergo partial fusion, resulting in less distinct inter-spherical boundaries and a reduction in the fine surface texture of the layers compared to LCA-2-700. This structural coarsening was consistent with the notable decrease in specific surface area and pore volume observed for LCA-2-800 (Table S2), likely due to partial collapse and sintering of the carbon framework at the elevated temperature. Based on the data summarized in Table S1, it could be inferred that both a fully developed spherical morphology and a smooth layered structure were crucial for enhancing the energy storage performance of the material.

In combination with the energy-dispersive X-ray spectroscopy (EDS) mapping of the carbon aerogel derived from sodium alginate and PA (Fig. S2), it could be observed that at 700 °C, the carbon aerogel formed from these precursors exhibits a distinct spherical architecture. As shown in Fig. S2, carbon aerogels lacking magnesium lignosulphonate exhibited no presence of sulphur, confirming that the sulphur in carbon aerogels originates from magnesium lignosulphonate. Furthermore, phytate servesd as the sole source of phosphorus in the raw materials forming the carbon aerogels. These carbon spheres were enriched with substantial amounts of phosphorus and oxygen, along with a comparatively low content of carbon. This phenomenon suggested that the spherical carbon structure effectively confined and dispersed the P and O elements. By incorporating sodium alginate and PA into the carbon aerogel system, we successfully fabricated the lignin-based carbon aerogel denoted as LCA-2-700. Based on the systematic comparison of the aforementioned morphology, structure, and electrochemical performance, it could be concluded that LCA-2-700 exhibited the

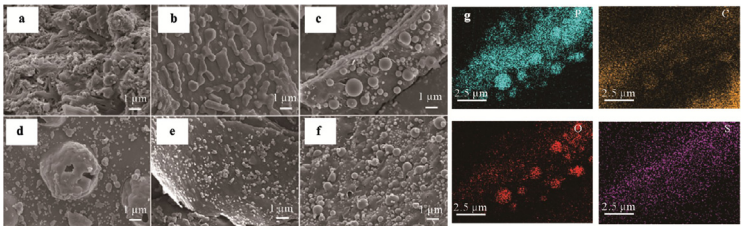

Fig. 3. Field-emission scanning electron microscopy (FE-SEM) images of (a) lignin-based aerogel with the LS:SA mass ratio of 5:5 and carbonized at 700 °C (LCA-700), (b) lignin-based aerogel with the LS:SA:PA mass ratio of 5:5:1 and carbonized at 700 °C (LCA-1-700), (c) LCA-2-700, (d) lignin-based aerogel with the LS:SA:PA mass ratio of 5:5:3 and carbonized at 700 °C (LCA-3-700), (e) LCA-2-600, and (f) LCA-2-800; (g) Energy-dispersive X-ray spectroscopy (EDS) image of LCA-2-700.

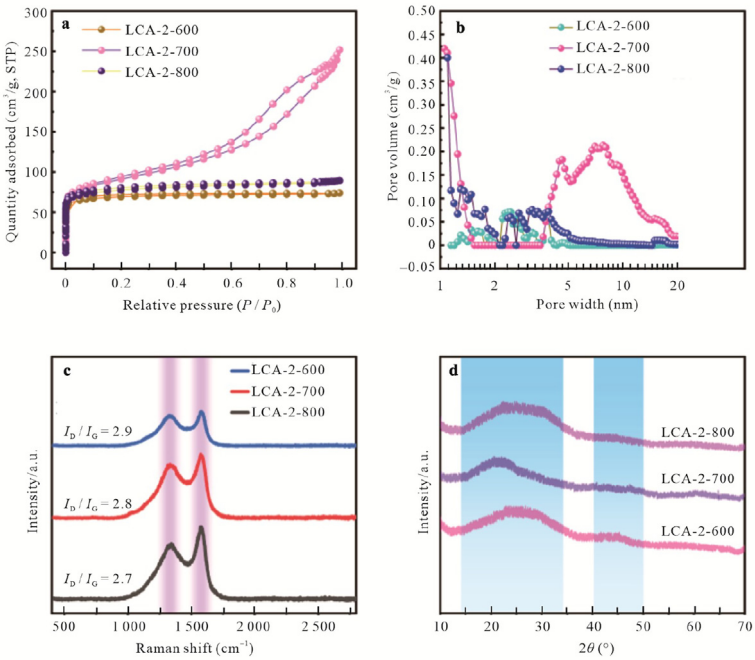

Fig. 4. (a) Nitrogen adsorption-desorption isotherms; (b) Pore size distribution curves; (c) Raman spectra; (d) X-ray diffraction (XRD) spectra of LCA-2-600, LCA-2-700 and LCA-2-800.
*I*D, the intensity of D-band; *I*G, the intensity of G-band.

most optimal combination of properties among all samples. It possessed the most regular spherical hierarchical structure, the largest specific surface area (273.107 m^2/g). Therefore, LCA-2-700 was subsequently focused on to further investigate its pore structure, surface chemistry, and charge storage mechanism. Fig. 3g displays EDS elemental mapping of LCA-2-700, illustrating the spatial distribution of various elements. The results indicated that oxygen (21.26%), phosphorus (3.35%), and sulfur (0.47%) were uniformly distributed across the surface of the carbon spheres. Such a structure significantly increased the accessible surface area for heteroatom dispersion compared to a layered morphology (Table S2). During electrochemical processes, this architecture introduced enhanced pseudocapacitive contributions by providing abundant active sites, thereby improving the overall material performance and achieving the goal of high-efficiency heteroatom doping.

The pore structure of the carbon aerogels was further characterized by N_2 adsorption-desorption measurements as shown in Figs. 4a and 4b. The isotherms of LCA-2-600, LCA-2-700, and LCA-2-800 all exhibited IV-type curves. A pronounced uptake at low relative pressure ($P/P_0 < 0.01$) suggested the presence of micropores in all three samples. Notably, within the medium relative pressure range ($0.4 < P/P_0 < 0.9$), LCA-2-700 showed an H3-type hysteresis loop, indicating the existence of a uniform mesoporous structure with a pore size distribution centered between 2–10 nm. As summarized in Table S3, the specific surface areas of LCA-2-600, LCA-2-700, and LCA-2-800 are 170.117, 273.107, and 196.835 m^2/g, respectively, with corresponding average pore sizes of 1.69 nm, 4.74 nm, and 1.86 nm, respectively. While LCA-2-600 and LCA-2-800 were predominantly microporous, LCA-2-700 exhibited a mesoporous-dominated structure, which was more conducive to electrochemical energy storage applications (Liu et al., 2016; Xiao et al., 2020; Kong and Wang, 2021).

As shown in Fig. 4c, all samples exhibited two characteristic peaks located at approximately 1350 (D band) and 1580 cm^{-1} (G band). The D band is associated with structural defects and disordered carbon domains, while the G band corresponds to the in-plane vibration of sp^2-hybridized graphitic carbon. The intensities of the G and D bands were determined by Gaussian fitting, and the degree of graphitization was evaluated based on the intensity ratio (I_D/I_G, where I_D is the intensity of D-band, I_G is the intensity of G-band) (Zhao et al., 2015; Kong and Wang, 2021). The I_G/I_D values for LCA-2-600, LCA-2-700, and LCA-2-800 were calculated to be 2.9, 2.8, and 2.7, respectively, indicating a slight increase in graphitization with the rise of temperature, although the overall change remained moderate. The structural properties of the samples were further investigated by X-ray diffraction (XRD). As depicted in Fig. 4d, the XRD patterns displayed broad and weak diffraction peaks at around 23° and 43.8°, corresponding to the (002) and (100) crystal planes of carbon, respectively, suggesting a predominantly amorphous structure.

The XPS survey spectrum of LCA-2-700 (Fig. 5a) confirmed the presence of carbon (C), oxygen (O), phosphorus (P), and sulfur (S). The high-resolution C 1s spectrum (Fig. 5b) could be deconvoluted into three components. The peak at 284.7 eV was assigned to C–C bonds in aliphatic/aromatic structures, the peak at 286.5 eV was attributed to C–O–C and C–O–P bonds, and the peak at 288.3 eV corresponded to C=O bonds. These results provided evidence for the formation of P–O–C bonds through dehydration between phosphate groups in PA and hydroxyl groups in sodium alginate. The O 1s spectrum (Fig. 5c) was fitted into three con-

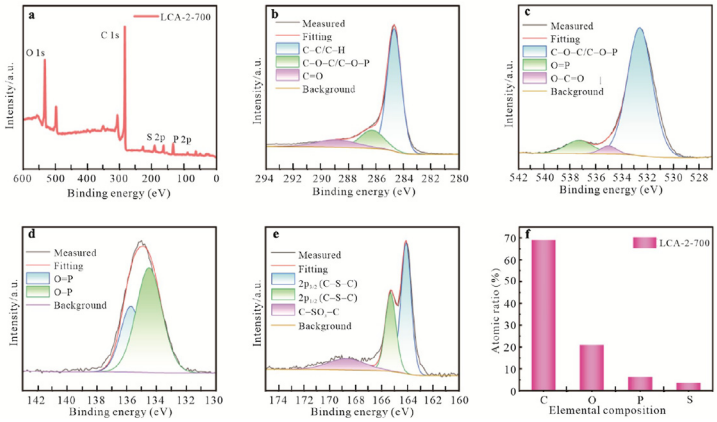

Fig. 5. (a) Full-survey X-ray photoelectron spectroscopy (XPS) spectra of LCA-2-700; High-resolution XPS spectra of the (b) C 1s, (c) O 1s, (d) P 2p, and (e) S 2p; (f) Elemental contents of LCA-2-700.

tributions: the peaks at 537.3 eV, 534.6 eV, and 532.6 eV were assigned to P=O, O–C=O, and C–O–C/C–O–P functional groups, respectively.

The P 2p spectrum in Fig. 5d could be deconvoluted into two components, i.e., P–O with binding energy of 134.6 eV and P=O with binding energy of 136.1 eV. Similarly, the S 2p spectrum in Fig. 5e could be fitted into three contributions: S $2p_{3/2}$ (C–S–C) at binding energy of 164.1 eV, S $2p_{1/2}$ (C–S–C) at binding energy of 165.4 eV, and oxidized sulfur species (–C–SO_x–C–) at binding energy of 168.7 eV. As summarized in the bar chart in Fig. 5f and Table S3, the elemental composition of LCA-2-700 consisted of C (69.1%), O (21.0%), P (6.3%), and S (3.6%). Combined with FT-IR results, the XPS data confirmed that the phosphate groups from PA promoted the dehydration and carbonization process during aerogel formation, facilitating the development of carbon layers. The identified P=O, P–O, and C–S–C functional groups contributed significantly to pseudocapacitance through reversible faradaic reactions in the alkaline electrolyte. Specifically, the phosphorus-containing groups (P=O/P–O) underwent reversible protonation/deprotonation processes, while the thiophene-type sulfur (C–S–C) enhanced electron delocalization and participated in quinone-hydroquinone-like redox reactions.

The exceptional performance of LCA-2-700 stemmed from the synergistic cooperation between its hierarchical pore structure and heteroatom functionality. The mesopore-dominated architecture (average pore size of 4.74 nm) provided efficient pathways for rapid ion diffusion, ensuring timely access of electrolyte ions to active sites. Meanwhile, the uniformly distributed P/S heteroatoms created abundant electrochemically active centers on the carbon surface. This synergy enabled both efficient electric double-layer formation in the mesopores and enhanced pseudocapacitance through surface redox reactions, overcoming the kinetic limitations typically encountered in microporous carbon materials. The electrochemical performance was then evaluated.

Electrochemical tests were also conducted in 1 mol/L H_2SO_4 to evaluate performance in acidic media, where the protonation of P and S functional groups could be favourable. However, the specific capacitances measured in H_2SO_4 were lower than those in KOH across all current densities, as shown in Fig. S3. Therefore, KOH was selected as the electrolyte for all subsequent tests.

The CV curves of LCA-2-600, LCA-2-700, and LCA-2-800 at the scanning rate of 20 mV/s were shown in Fig. 6a. Each CV curve presented a wide peak in the potential range from –0.6 to –0.8 V, indicating that the induced heteroatoms brought pseudo-capacitance behaviour, and the reversible redox reaction related to the doped functional groups of heteroatoms was an observed pseudo-capacitance characteristic, which was the result of the co-existence of pseudocapacitance and double-layer capacitance (Cheng et al., 2020). As the activation temperature increased from 600 to 700 °C, the area under the CV curve (a measure of electrochemical energy storage capacity) increased significantly. However, further increasing the activation temperature to 800 °C resulted in a decrease in the area under the CV curve. As a result, LCA-2-700 showed the best energy storage performance in the test samples.

As depicted in Fig. 6b, the constant-current charge-discharge (GCD) curves of LCA-2-600, LCA-2-700, and LCA-2-800 at a current density of 1 A/g tended to form isosceles triangles, signifying the synergistic contribution of both double-layer capacitance (EDLC) and pseudocapacitive processes (Cao et al., 2020). The energy storage results obtained from the GCD test were in agreement with those obtained from the CV tests. The specific capacitance of LCA-2-600, LCA-2-700, and LCA-2-800 at a current density of 1 A/g is 131, 335, and 254 F/g, respectively. In addition, as shown in Fig. 6c, the capacitance retention of LCA-2-700 reached 77.3%, demonstrating good electrochemical stability. It still showed excellent magnification performance even at a current density of 10 A/g because of the pseudocapacitance brought about by heteroatom doping. This doping strategy effectively optimized the electrochemical properties of the material, enabling it to maintain a high capacitance value and good cycle stability during the high-current charge-discharge process, further proving the potential application value of LCA-2-700 in high-performance energy storage devices.

Figs. 6d and 6e are CV curves at different sweep speeds and GCD curves at different current densities, respectively. Under cyclic voltammetry (CV) tests at different sweep speeds, the CV curve of LAC-2-700 presented a rectangular shape, and this shape could remain stable under various sweep speeds, which fully indicated that LAC-2-700 had excellent electrochemical reversibility. The GCD curves of LAC-2-700 showed excellent coulomb efficiency in GCD tests at different current densities. At the current density of 0.5 A/g, the specific capacitance reached 362 F/g.

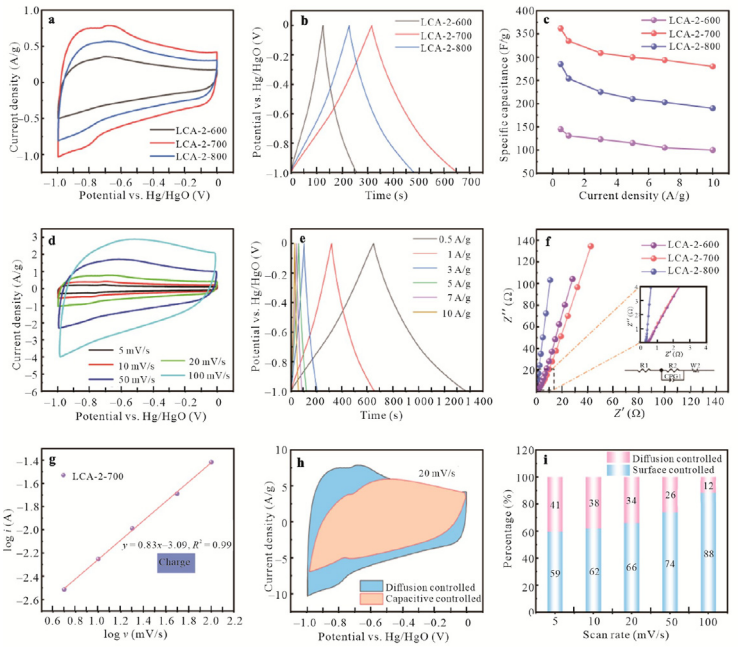

Fig. 6. Electrochemical analysis (three-electrode system). (a) Cyclic voltammetry (CV) curves at 20 mV/s (Potential window range from –1.0 to 0 V); (b) Galvanostatic charge-discharge (GCD) curves at current density of 1 A/g; (c) Rate performance; (d) The CV curves at different scanning rates (potential window range from –1.0 to 0 V); (e) The GCD curves at different current densities of LAC-2-700; (f) Electrochemical impedance spectroscopy (EIS) curves of LAC-2-600, LAC-2-700, and LAC-2-800; (g) The log *v* vs. log i for LCA-2-700; (h) Typical charge storage contributions from capacitive and diffusive from the CV curve at the specific scan rate of 20 mV/s; (i) Capacitive and diffusive contributions at different scanning rates

i, current; *v*, scan rate; SCE, saturated calomel electrode.

In addition, electrochemical impedance spectroscopy (EIS) analysis was performed in the range of 10 mHz–100 kHz to gain an in-depth understanding of the charge transport mechanics of the electrode materials. As shown in Fig. 6f, the EIS curves of carbon aerogel for LCA-2-600, LCA-2-700, and LCA-2-800 consisted of two different parts: the semicircle of the high frequency region, which corresponded to the charge transfer resistance between electrolyte ions and electrons; a straight line in the low frequency region, indicating the diffusion and migration of electrolyte ions within the porous structure. Through careful observation of the EIS curve, we could find that the LCA-2-700 line was closer to the 45 ˚C angle in the low frequency region, which clearly showed its excellent ion diffusion performance. In addition, after fitting the impedance of LCA-2-700 in the high frequency region, a very low value of 0.0045 Ω was obtained, which was undoubtedly conducive to the rapid charge transport. The impedance value, as a key parameter to evaluate the charge transfer characteristics of the material, further confirmed the excellent performance of LCA-2-700 in terms of electrochemical performance.

The CV curves of LCA-2-700, ranging from 5 to 100 mV/s, were depicted in Fig. 6d. These curves consistently demonstrated superior morphological stability, which was attributed to the rapid charge transfer kinetics at the electrode/electrolyte interface. The charge storage mechanism of LCA-2-700 was further elucidated by assessing the contributions of current to its surface and diffusion-controlled processes. The capacitive contribution of the LCA-2-700 electrode was quantified using the Dunn equation, and the parameter *b* (slope of the line in Fig. 6g) depicted in Fig. 6g characterized the electrochemical processes. A value of $b = 1$ signifies a predominantly surface-controlled capacitive behaviour, whereas $b = 0.5$ indicates a diffusion-controlled process, typically associated with Faradaic reactions (Huang et al., 2023; Liu et al., 2023). Values of *b* between 0.5 and 1.0 suggest a hybrid mechanism. The *b* values obtained from our analysis confirmed the mixed charge storage mechanism of LCA-2-700. As shown in Fig. 6i, the capacitance of LCA-2-700 showed a tendency to decrease with the increase of the scanning rate. At the same time, the contribution rate of pseudocapacitance storage to the entire thin film capacitance increased significantly from 59% to 88%, which clearly indicated that the pseudo-capacitance effect caused by heteroatom doping played a dominant role in the entire capacitance mechanism, and thus strongly confirmed the successful implementation of the heteroatom doping strategy.

The LCA-2-700 electrodes within a symmetric two-electrode configuration were employed for practical applications, utilizing LCA-2-700 for both the positive and negative electrode materials and a 6 mol/L aqueous potassium hydroxide solution as the electrolyte to construct a symmetric supercapacitor device (Fig. 7a). Fig. 7b presents the CV curves of the LCA-2-700//LCA-2-700 SSC across various voltage windows. Due to the successful doping of heteroatoms, at a scanning rate of 50 mV/s, the CV curves maintained a rectangular shape even when the voltage window was expanded from 1 V to 1.4 V. This indicates a broad potential window and the capability to deliver a higher energy density compared to systems with narrower windows.

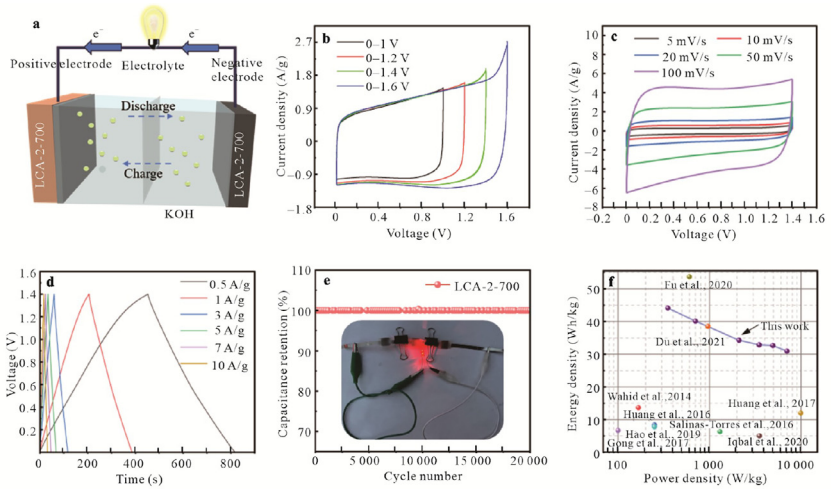

Fig. 7. Electrochemical analysis of symmetrical supercapacitors LCA-2-700//LCA-2-700. (a) Schematic diagram; (b) The CV curves under different voltage windows; (c) The CV curves at different scanning rates; (d) The GCD curves at different current densities; (e) Cycling performance at the current density of 10 A/g for 20,000 cycles and with light-emitting diode; (f) Ragone plots.

Fig. 7c depicts the CV curves at different scanning rates. The CV curves retained their rectangular shape without significant distortion as the scanning rate increased from 5 to 100 mV/s, signifying the LCA-2-700 electrode material's excellent capacitive behaviour and reversibility. Fig. 7d illustrates the GCD curves at various current densities, revealing a specific capacitance of 147.1 F/g at the current density of 1 A/g. Upon calculation, the energy density at this current density was determined to be 40.1 W·h/kg, corresponding to a power density of 700 W/kg.

Fig. 7e displays the capacitance retention alongside the corresponding light-emitting diode plot, demonstrating that the capacitance retention of the symmetric two-electrode system remained at 82.5% after 20,000 cycles. This underscores the system's outstanding cyclability, positioning the LCA-2-700 samples as promising candidates for electrochemical energy storage electrodes. Fig. 7f presents the Ragone plot of the constituent devices. It highlighted that the LCA-2-700//LCA-2-700 SSC exhibited superior energy and power densities, benefitted from its expanded electrochemical reaction window. These values significantly surpassed the energy density of the previous reported lignin-based carbon supercapacitor devices with a two-electrode system (Wahid et al., 2014; Huang et al., 2016; Salinas-Torres et al., 2016; Gong et al., 2017; Huang et al., 2017; Hao, et al., 2019; Fu et al., 2020; Du et al., 2021).

4. Conclusions

This study presented a simple and efficient strategy for synthesizing phosphorus/sulfur (P/S) co-doped carbon aerogels using phytic acid (PA), magnesium lignosulfonate (LS), and sodium alginate (SA) as sustainable precursors. The innovative approach leveraged the inherent sulfur in magnesium lignosulfonate for in-situ self-doping and utilized PA as both a phosphorus source and a morphology regulator. We have demonstrated that the superior electrochemical performance originated from the synergistic interplay between the spherical mesoporous architecture and P/S dual-doping. The mesoporous framework facilitated ion transport kinetics, while the heteroatom functionalities provided substantial pseudocapacitance through specific redox reactions involving P=O/P–O and C–S–C groups in alkaline electrolyte. Performance testing confirmed that the optimized LCA-2-700 sample exhibited a remarkable specific capacitance of 362 F/g at the current density of 0.5 A/g. The assembled symmetrical supercapacitor achieved a high energy density of 40.1 W·h/kg at a power density of 700 W/kg and maintains 82.5% capacitance retention after 20,000 cycles. The high performance is attributed to the synergistic effect of the spherical structure and P/S dual-heteroatom doping. This work not only demonstrates the high-value utilization of biomass waste but also provides a promising electrode material for advanced supercapacitors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jobab.2026.100234.

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